



# MANAGERIAL ACCOUNTING (135)

**State – 2020**

**Multiple Choice & Short Answer:**

Multiple Choice (15 @ 2 points) \_\_\_\_\_ (30 points)

Short Answers (10 @ 1 point) \_\_\_\_\_ (10 points)

**Production:**

Problem 1: Variable Income Statement \_\_\_\_\_ (28 points)

Problem 2: Budgets \_\_\_\_\_ (35 points)

Problem 3: Break Even Analysis (5 @ 3 points) \_\_\_\_\_ (15 points)

***TOTAL POINTS*** \_\_\_\_\_ ***(118 points)***

**Judge/Graders: Please double check and verify all  
scores and answer keys!**

Property of Business Professionals of America.  
May be reproduced only for use in the Business Professionals of America  
*Workplace Skills Assessment Program* competition.

### General Instructions

You have been hired as a Financial Assistant and will be keeping the accounting records for Professional Business Associates, located at 5454 Cleveland Avenue, Columbus, Ohio 43231. Professional Business Associates provides accounting and other financial services for clients. You will complete jobs for Professional Business Associates' own accounting records, as well as for clients.

You will have 90 minutes to complete your work. The test is divided into two parts. The multiple choice and short answer questions should be completed first, and then the accounting problems may be completed in any order.

Your name and/or school name should **not** appear on any work you submit for grading. Write your contestant number in the provided space. Staple all pages in order before you turn in your test.

Assumptions to make when taking this assessment:

Percents rounded to two decimal places or the tenth percent (Ex: 22.8%)

#### Multiple Choice (2 points each)

- |      |       |       |
|------|-------|-------|
| 1. A | 6. A  | 11. A |
| 2. D | 7. D  | 12. B |
| 3. A | 8. B  | 13. C |
| 4. D | 9. D  | 14. C |
| 5. C | 10. B | 15. A |

#### Short Answer (1 point each)

- |       |       |
|-------|-------|
| 16. V | 21. V |
| 17. V | 22. F |
| 18. M | 23. F |
| 19. F | 24. V |
| 20. V | 25. V |

**Graders – 2 points for each bolded title or number for a total of 28 points.**

DuKray Concrete Company reported the following data for the month of March 2020. Beginning and ending inventories were negligible so were not considered. Prepare a variable costing Income Statement in the format provided.

|                                                                                    | Total Cost           | Number of Units | Unit Cost       |
|------------------------------------------------------------------------------------|----------------------|-----------------|-----------------|
| Manufacturing Costs:                                                               |                      |                 |                 |
| Variable                                                                           | \$ 288,000           | 20,000          | \$ 54.40        |
| Fixed                                                                              | <u>512,000</u>       | <u>20,000</u>   | <u>25.60</u>    |
| Total                                                                              | <u>\$ 800,000</u>    |                 | <u>\$ 80.00</u> |
|                                                                                    |                      |                 |                 |
| Selling & Admin Expenses:                                                          |                      |                 |                 |
| Variable                                                                           | \$3.50 per unit sold |                 |                 |
| Fixed                                                                              | \$ 125,000           |                 |                 |
|                                                                                    |                      |                 |                 |
| <b>In the month of March, 18,000 units were sold at a price of \$120 per unit.</b> |                      |                 |                 |

|                                                                                            |                       |                          |
|--------------------------------------------------------------------------------------------|-----------------------|--------------------------|
| DuKray Concrete Company<br>Variable Income Statement<br>For the month ended March 30, 2020 |                       |                          |
| <b>Sales</b>                                                                               |                       | <b>\$ 2,160,000</b>      |
| <b>Variable Manufacturing</b>                                                              |                       | <b>( 979,200)</b>        |
| Manufacturing Margin                                                                       |                       | 1,180,800                |
| <b>Variable Selling &amp; Admin</b>                                                        |                       | <b>( 63,000)</b>         |
| <b>Contribution Margin</b>                                                                 |                       | <b>1,117,800</b>         |
| Fixed Costs:                                                                               |                       |                          |
| <b>Manufacturing</b>                                                                       | <b>512,000</b>        |                          |
| <b>Selling &amp; Admin</b>                                                                 | <b><u>125,000</u></b> | <b>( 637,000)</b>        |
|                                                                                            |                       |                          |
| <b>Net Income</b>                                                                          |                       | <b><u>\$ 480,800</u></b> |
|                                                                                            |                       |                          |
|                                                                                            |                       |                          |
|                                                                                            |                       |                          |

Problem 2 – Budgets (35 points)**Graders – 1 point for each bolded number for a total of 35 points.**

Control Production requests your assistance in preparing budgets for the next year. Use the information below and formats on the next page to prepare a Sales Budget, Production Budget, and Direct Materials Budget.

- a. Estimated sales of the Small and Large Controls for the North, East, West, and South Divisions are as follows:

## North Division

|                |                              |
|----------------|------------------------------|
| Small Controls | 1,500 units at \$40 per unit |
| Large Controls | 4,500 units at \$65 per unit |

## East Division

|                |                              |
|----------------|------------------------------|
| Small Controls | 2,000 units at \$50 per unit |
| Large Controls | 6,000 units at \$75 per unit |

## West Division

|                |                              |
|----------------|------------------------------|
| Small Controls | 6,000 units at \$60 per unit |
| Large Controls | 3,000 units at \$80 per unit |

## South Division

|                |                              |
|----------------|------------------------------|
| Small Controls | 1,000 units at \$55 per unit |
| Large Controls | 2,200 units at \$70 per unit |

- b. Estimated beginning inventories and desired ending inventories:

|                   | <b>Beginning Inventory</b> | <b>Ending Inventory</b> | <b>Used in Production</b> | <b>Direct Mat Purchase Price</b> |
|-------------------|----------------------------|-------------------------|---------------------------|----------------------------------|
| Direct Materials: |                            |                         |                           |                                  |
| Casing            | 280 units                  | 300 units               |                           |                                  |
| Wire              | 5,410 feet                 | 4,500 feet              |                           |                                  |
|                   |                            |                         |                           |                                  |
| Small Control     |                            |                         |                           |                                  |
| Casing            |                            |                         | 1 per unit                | \$9.10 per unit                  |
| Wire              |                            |                         | 20 feet                   | \$0.10 per foot                  |
| Large Control     |                            |                         |                           |                                  |
| Casing            |                            |                         | 1 per unit                | \$9.10 per unit                  |
| Wire              |                            |                         | 40 feet                   | \$0.10 per foot                  |
|                   |                            |                         |                           |                                  |
| Finished Goods:   |                            |                         |                           |                                  |
| Small Controls    | 280                        | 310                     |                           |                                  |
| Large Controls    | 330                        | 350                     |                           |                                  |

| Control Productions<br>Sales Budget<br>For the year ending December 31, 2021 |                   |                    |                     |
|------------------------------------------------------------------------------|-------------------|--------------------|---------------------|
| Product and Area                                                             | Unit Sales Volume | Unit Selling Price | Total Sales         |
| Small Controls:                                                              |                   |                    |                     |
| North                                                                        | 1,500             | \$ 40              | \$ 60,000           |
| East                                                                         | 2,000             | 50                 | 100,000             |
| West                                                                         | 6,000             | 60                 | 360,000             |
| South                                                                        | <u>1,000</u>      | 55                 | <u>55,000</u>       |
|                                                                              | <b>10,500</b>     |                    | <b>\$ 575,000</b>   |
|                                                                              |                   |                    |                     |
| Large Controls:                                                              |                   |                    |                     |
| North                                                                        | 4,500             | 65                 | 292,500             |
| East                                                                         | 6,000             | 75                 | 450,000             |
| West                                                                         | 3,000             | 80                 | 240,000             |
| South                                                                        | <u>2,200</u>      | 70                 | <u>154,000</u>      |
|                                                                              | <b>15,700</b>     |                    | <b>1,136,500</b>    |
|                                                                              |                   |                    |                     |
| <b>Total Sales</b>                                                           |                   |                    | <b>\$ 1,711,500</b> |
|                                                                              |                   |                    |                     |

| Control Productions<br>Production Budget<br>For the year ending December 31, 2021 |                |                |
|-----------------------------------------------------------------------------------|----------------|----------------|
|                                                                                   | Units          |                |
|                                                                                   | Small Controls | Large Controls |
| Sales                                                                             | <b>10,500</b>  | <b>15,700</b>  |
| + Desired Ending Inventory                                                        | <b>310</b>     | <b>350</b>     |
|                                                                                   | <b>10,810</b>  | <b>16,050</b>  |
| - Beginning Inventory                                                             | <b>(280)</b>   | <b>(330)</b>   |
| Units to Produce                                                                  | <b>10,530</b>  | <b>15,720</b>  |
|                                                                                   |                |                |
|                                                                                   |                |                |
|                                                                                   |                |                |
|                                                                                   |                |                |

| Control Productions<br>Direct Materials Purchases Budget<br>For the year ending December 31, 2021 |                   |                  |                   |
|---------------------------------------------------------------------------------------------------|-------------------|------------------|-------------------|
|                                                                                                   | Casings           | Wire             | Total             |
| Units to Produce:                                                                                 |                   |                  |                   |
| Small                                                                                             | 10,530            | *210,600         |                   |
| Large                                                                                             | 15,720            | **628,800        |                   |
| + Desired Ending Inventory                                                                        | <u>300</u>        | <u>4,500</u>     |                   |
|                                                                                                   | 26,550            | 843,900          |                   |
| - Beginning Inventory                                                                             | <u>-280</u>       | <u>-5,410</u>    |                   |
| Total Units to Purchase                                                                           | 26,270            | 838,490          |                   |
| * Unit Price                                                                                      | <u>* \$ 9.10</u>  | <u>* \$0.10</u>  |                   |
| Total Direct Mat Purchases                                                                        | <u>\$ 239,057</u> | <u>\$ 83,849</u> | <u>\$ 322,906</u> |
|                                                                                                   |                   |                  |                   |
|                                                                                                   |                   |                  |                   |
|                                                                                                   |                   |                  |                   |
|                                                                                                   |                   |                  |                   |
|                                                                                                   |                   |                  |                   |
|                                                                                                   |                   |                  |                   |

\* 10,530 units x 20 feet = 210,600 feet

\*\* 15,720 units x 40 feet = 628,800 feet

**Grader – 3 points for each bolded number for a total of 15.**

Complete the following Break-Even situations. Each option should start with the original assumptions  
(Round each answer to the nearest whole unit):

Unit Selling Price        \$1,200

Unit Variable Cost        800

Total Fixed Costs        415,000

1. Calculate the Break-Even point in units.

**1,038 units**

2. Assume the company is considering paying \$2,000 more per year to each of its **four** factory supervisor's salaries. What is the new Break-Even point in units?

**1,058 units**

3. What is the new Break-Even point be if cost of direct materials increased by \$50 per unit?

**1,186 units**

4. What would be the new Break-Even point is the selling price increased to \$1,280?

**865 units**

5. What is the sales volume in units needed to earn a target profit of \$500,000?

**2,288 units**